SOFTWARE REQUIREMENTS SPECIFICATION

for

SPORTS RESOURCE MANAGEMENT SYSTEM

Version 1.0

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1 Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to describe the functional and non-functional requirements of the Sports Resource Management System. The system is being developed as a mobile application to organize sports equipment, facility availability, equipment requests, and facility bookings within a university campus.

This document defines the project's scope, objectives, stakeholders, user classes, major system features, operating environment, constraints, and requirements. It is intended to guide implementation and testing and to provide a common understanding of the system among team members and evaluators.

The application is developed using Flutter and Dart. A Node.js backend handles server-side operations, while PostgreSQL is used for persistent data storage.

The current submission describes a working prototype. Administrator functionality, the maintenance staff interface, feedback, and notifications are not fully functional and are treated as planned extensions rather than completed features.

1.2 Intended Audience and Reading Suggestions

This document is intended for:

- Project developers responsible for implementing the application.
- Project team members coordinating frontend, backend, and database work.
- Faculty members evaluating the software engineering project.
- Testers verifying functional and non-functional requirements.
- Future maintainers extending the application.

Chapter 1 introduces the project, its purpose, scope, objectives, and stakeholders. Chapter 2 describes the overall system, users, operating environment, assumptions, and constraints. Chapter 3 defines system features and functional requirements. Chapter 4 describes non-functional requirements and quality attributes. Chapter 5 lists other requirements and future extensions. Relevant diagrams will be added to the designated sections after they are prepared.

1.3 Project Scope

The Sports Resource Management System is a mobile application intended to make university sports resource management more organized and accessible.

The system supports students or players, coaches, sports officers, administrators, and maintenance staff. Each role has different responsibilities and access to functionality.

The main scope of the system includes:

- User role selection and login.

- Viewing sports equipment and its availability.
- Requesting available equipment.
- Viewing sports facilities.
- Checking facility availability.
- Booking available facility time slots.
- Managing equipment and facilities through the Sports Officer role.
- Storing user, equipment, facility, booking, and request information.
- Providing a foundation for maintenance tracking, feedback, notifications, and administration.

Facility bookings are intended to be confirmed automatically when the selected time slot is available. The Sports Officer does not need to approve each normal facility booking.

Students and coaches can request available equipment directly without waiting for a Sports Officer to approve the request. Equipment issue and return records may be maintained as part of the resource management process.

The application is mobile-only. A separate web-based administration dashboard is outside the current project scope.

Administrator functionality, the maintenance staff interface, feedback, and notifications are included in the system design but are not fully functional in the current prototype.

1.4 Objectives

The main objectives of the project are:

1. To develop a centralized mobile application for managing university sports equipment and facilities.
2. To provide role-based access for students, coaches, sports officers, administrators, and maintenance staff.
3. To allow students and coaches to view equipment availability and request available equipment.
4. To allow users to view sports facilities and book available time slots.
5. To automatically confirm facility bookings when the requested slot is available.
6. To maintain structured records of users, equipment, facilities, bookings, and equipment requests.
7. To support identification of equipment through unique Equipment IDs.
8. To provide a foundation for maintenance tracking, feedback, notifications, and system administration.
9. To apply software engineering principles such as requirement analysis, modular development, database design, and testing.

1.5 Stakeholders

The stakeholders of the Sports Resource Management System are:

- **Students/Players:** Use the application to view sports resources, request equipment, and book facilities.
- **Coaches:** Use the application to access sports facilities, request equipment, and book facilities for sports activities.
- **Sports Officer:** Manages equipment and facilities and handles operational resource management.
- **Maintenance Staff:** Intended to handle equipment repair records and update maintenance status.
- **Administrator:** Intended to manage system-level users, roles, and administrative information.
- **Project Team:** Designs, implements, tests, and maintains the application.
- **Project Supervisor/Evaluator:** Reviews the requirements, design, implementation, and prototype progress.

1.6 Definitions, Acronyms, and Abbreviations

SRS Software Requirements Specification.

UI User Interface.

API Application Programming Interface.

DBMS Database Management System.

Flutter Cross-platform application development framework.

Dart Programming language used by the Flutter application.

Node.js Runtime used for the backend server.

PostgreSQL Relational database management system.

Equipment ID Unique identifier for an equipment record.

Prototype An initial working implementation used to demonstrate core system functionality.

2 Overall Description

2.1 Product Perspective

The Sports Resource Management System is a mobile client-server application. It is intended to replace fragmented or manual handling of sports resource information with a centralized digital system.

The mobile application is developed using Flutter and Dart. It communicates with a Node.js backend, which handles application requests and accesses the PostgreSQL database.

The database stores information about users, equipment, facilities, bookings, equipment requests, maintenance, feedback, and notifications. Some of these entities support features that are still under development.

The system is divided into three major technical components:

1. **Mobile application:** Presents screens and controls through which users access the system.
2. **Backend:** Processes application requests and performs server-side operations.
3. **Database:** Stores and retrieves persistent system data.

2.1.1 System Architecture Diagram

The system architecture diagram will illustrate the relationship between the Flutter mobile application, Node.js backend, and PostgreSQL database.

2.2 User Classes and Characteristics

The system is designed for the following user classes.

2.2.1 Student/Player

Students or players are the primary users of the resource-access features. They can log in, view equipment information, request available equipment, view facilities, and book available facility time slots.

2.2.2 Coach

Coaches can view sports facilities, check availability, request equipment, and book available facilities for sports activities. Their core resource-access features are similar to those of students.

2.2.3 Sports Officer

The Sports Officer is responsible for operational management of sports resources. The intended functions include managing equipment and facilities and handling equipment issue and return information.

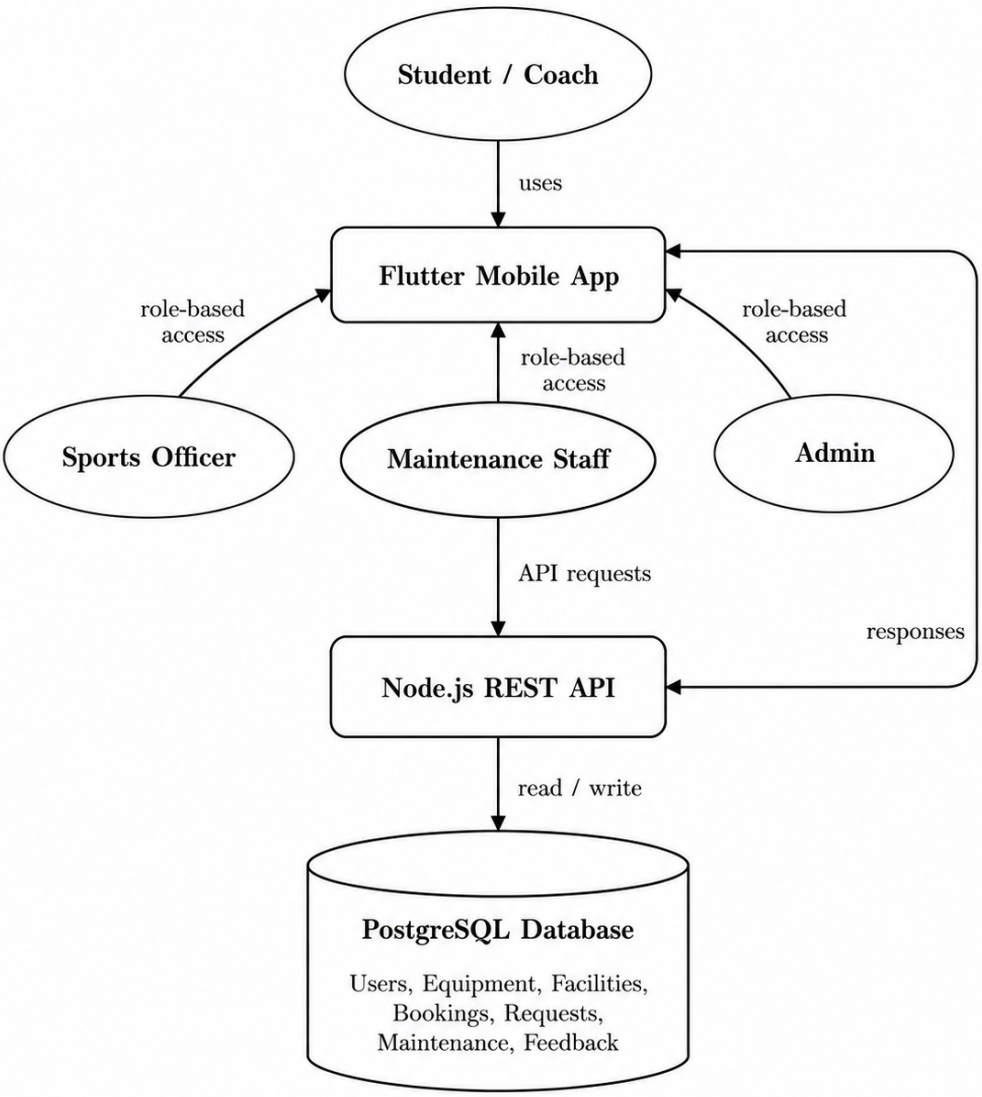


Figure 2.1: System Architecture Diagram

Figure 2.1: System Architecture of the Sports Resource Management System

2.2.4 Maintenance Staff

Maintenance staff are intended to access equipment maintenance information, identify equipment by Equipment ID, and update repair status. The maintenance interface is not fully functional in the current prototype.

2.2.5 Administrator

The administrator role is intended for system-level user and role management. This functionality is not fully functional in the current prototype.

2.2.6 User Classes Diagram

A diagram showing the user classes and their major interactions.

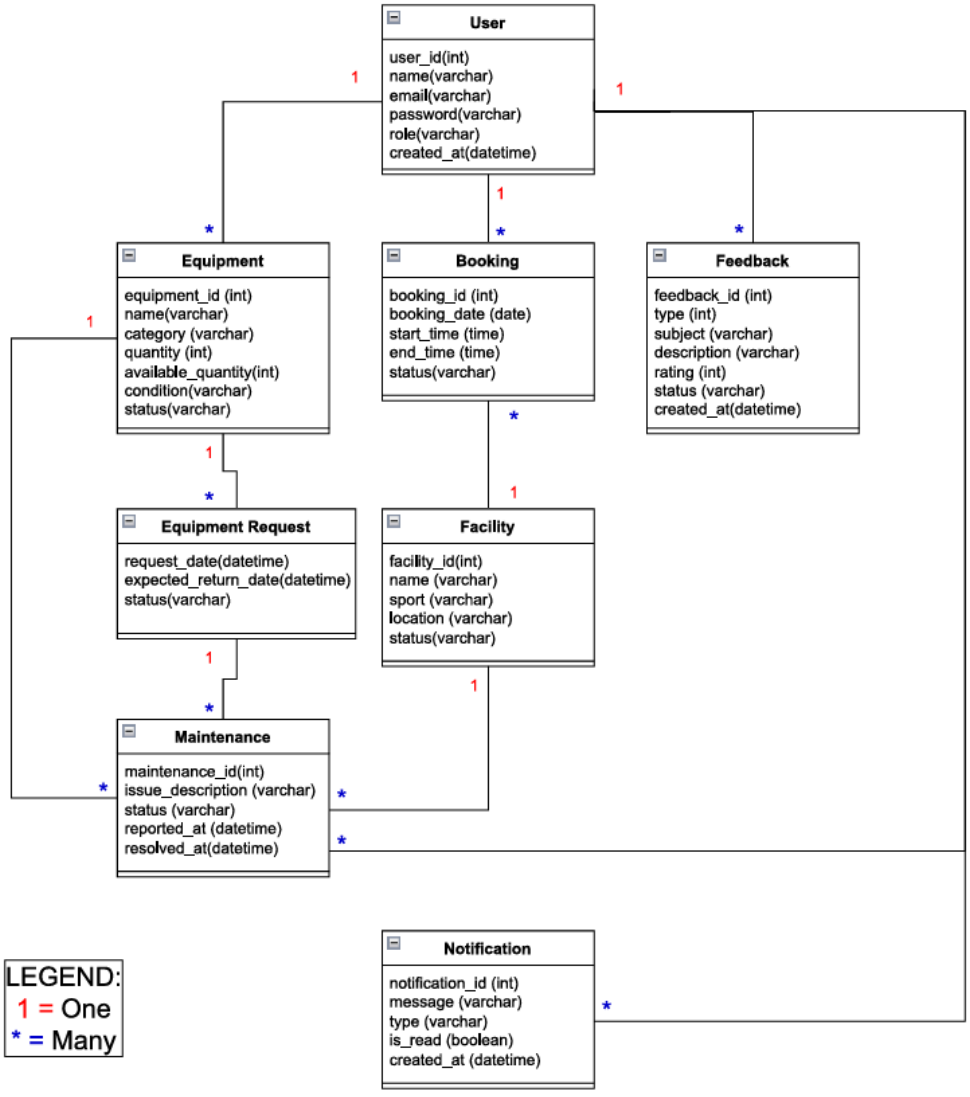


Figure 2.2: User Classes

2.3 Product Functions

The main functions of the system are described below.

- **Authentication and role selection:** Allows a user to select their role and access the corresponding login workflow.
- **Equipment browsing:** Displays available equipment and relevant information such as category, quantity, condition, and status.
- **Equipment requests:** Allows students and coaches to request available equipment directly without a separate approval step.
- **Facility browsing:** Displays facility information, including name, sport, location, and status.
- **Facility booking:** Allows users to select a facility, date, and time slot. The system checks availability and automatically confirms a booking if the slot is available.
- **Equipment and facility management:** Provides the Sports Officer with functionality for managing resources.
- **Maintenance tracking:** Planned functionality for recording equipment repair information and status.
- **Feedback:** Planned functionality for collecting suggestions, ratings, and complaints.
- **Notifications:** Planned functionality for communicating booking and equipment-related updates.
- **Administration:** Planned functionality for system-level user and role management.

2.4 Operating Environment

The system uses the following development and runtime environment:

- **Mobile framework:** Flutter.
- **Application language:** Dart.
- **Backend runtime:** Node.js.
- **Database:** PostgreSQL.
- **Development environment:** Visual Studio Code.
- **Version control:** Git and GitHub.

The mobile application is intended to run on supported mobile devices. The backend and database must be accessible to the application for operations requiring server-side processing or persistent data.

2.5 Design and Implementation Constraints

The project is subject to the following constraints:

- The application must remain a mobile application; a separate web dashboard is not part of the current scope.
- Flutter and Dart are used for mobile application development.

- Node.js is used for backend functionality.
- PostgreSQL is used for persistent relational data storage.
- The system must maintain consistent records for equipment, facilities, bookings, and requests.
- Facility booking must check for overlapping reservations before confirming a requested slot.
- Equipment availability must reflect stored quantities and status information.
- The current project is a working prototype, not a production deployment.
- Administrator, maintenance, feedback, and notification modules must not be represented as fully implemented until they are working.

2.6 Assumptions and Dependencies

The following assumptions apply to the proposed system:

- Users have access to a supported device on which the mobile application can run.
- Users provide valid login information.
- Sports equipment and facility information is entered and maintained accurately by authorized personnel.
- The backend and PostgreSQL database are available when server-side operations are required.
- Facility booking availability is determined from existing booking records stored in the system.
- Each equipment record has an identifier that can distinguish it from other equipment records.
- Features marked as under development will require further implementation and testing before they can be treated as complete.

2.7 Use Case Diagram

The Use Case Diagram will represent the major interactions between Students/Players, Coaches, Sports Officers, Maintenance Staff, and Administrators.

It will include the core functions of login, viewing equipment, requesting equipment, viewing facilities, booking facilities, and managing sports resources. Planned functionality such as maintenance, feedback, notifications, and administration will be identified appropriately.

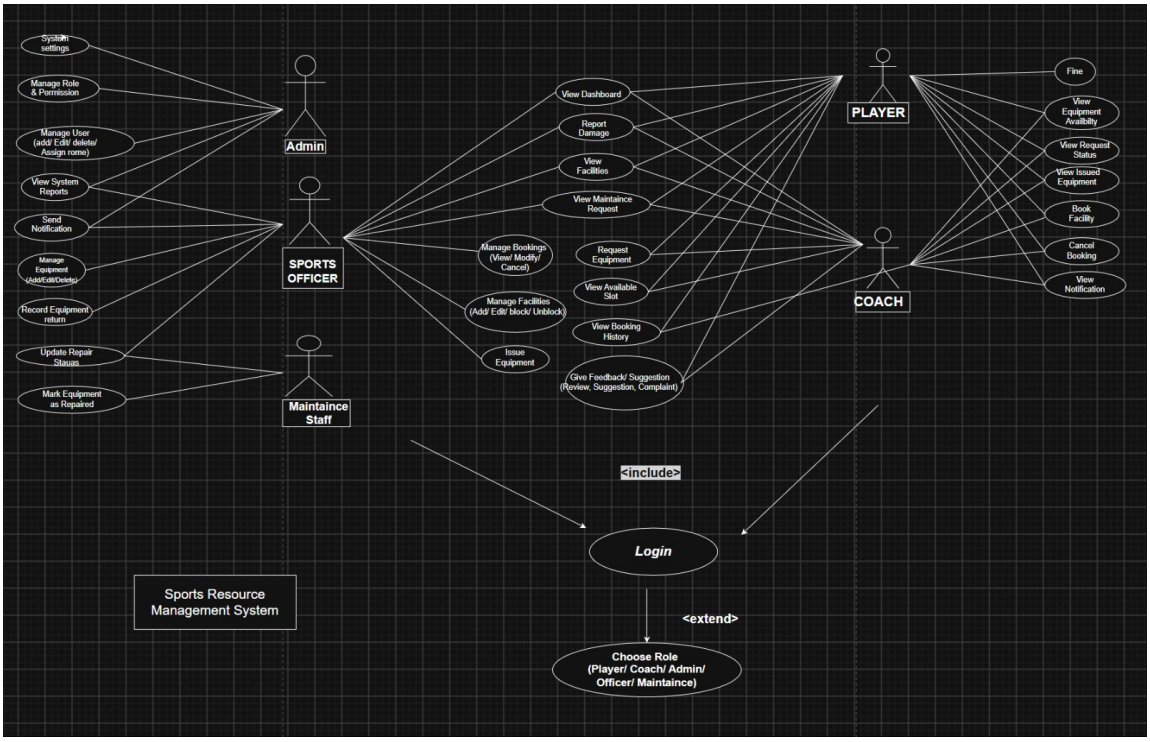


Figure 2.3: Use Case Diagram

3 System Features

3.1 Description and Priority

The system features are prioritized according to their importance for the working prototype.

1. **Authentication and role selection: High priority.** Users must be able to access the application through the login workflow appropriate to their role.
2. **Equipment browsing and requests: High priority.** Users must be able to view equipment information and request available equipment.
3. **Facility browsing and booking: High priority.** Users must be able to view facilities and book available time slots. Available slots are confirmed automatically.
4. **Equipment and facility management: High priority.** The Sports Officer must be able to maintain core resource information.
5. **Maintenance tracking: Planned.** Maintenance staff should be able to identify equipment and update repair information.
6. **Administrator functions: Planned.** The administrator should eventually be able to manage users and roles.
7. **Feedback and notifications: Planned.** These modules will provide additional communication and feedback functionality.

3.2 Functional Requirements

Functional requirements specify the behaviour that the system is expected to provide. Each requirement has a unique identifier to support implementation and testing.

3.2.1 Authentication and Role Selection

FR-01
The system shall allow a user to select their role before proceeding to the corresponding login workflow.

FR-02
The system shall provide a login interface for users.

FR-03
The system shall restrict role-specific operations to users authorized for those operations.

FR-04
The system shall display appropriate feedback when login information is invalid or an operation cannot be completed.

3.2.2 Equipment Browsing and Requests

FR-05

The system shall allow students and coaches to view sports equipment information.

FR-06

The system shall display relevant equipment details, including its name, category, quantity, availability, condition, and status where those details are available.

FR-07

The system shall allow students and coaches to request equipment that is available for use.

FR-08

The system shall check equipment availability before accepting a request.

FR-09

The system shall record equipment request information in the database.

FR-10

The system shall allow the relevant resource-management functionality to maintain equipment information.

FR-11

The system shall support recording equipment issue and return information where that functionality is implemented.

3.2.3 Facility Browsing and Booking

FR-12

The system shall allow students and coaches to view available sports facilities.

FR-13

The system shall display relevant facility details, including name, sport, location, and status where available.

FR-14

The system shall allow a user to select a facility and a requested date and time slot.

FR-15

The system shall check existing booking records to determine whether the requested time slot is available.

FR-16

The system shall automatically confirm a booking when the selected time slot is available.

FR-17

The system shall not confirm a booking when the selected time slot conflicts with an existing booking.

FR-18

The system shall store booking details, including the relevant user, facility, date, time, and booking status.

FR-19

The system shall allow users to view their booking information.

FR-20
The system shall support cancellation of bookings if cancellation functionality is enabled in the implemented application.

3.2.4 Sports Officer Functions

FR-21
The system shall allow the Sports Officer to manage equipment information.

FR-22
The system shall allow the Sports Officer to manage facility information.

FR-23
The system shall allow the Sports Officer to view equipment requests and relevant resource information.

FR-24
The system shall support updates to equipment quantity, availability, condition, and status where applicable.

3.2.5 Maintenance Functions

The following requirements describe planned functionality and are not fully implemented in the current prototype.

FR-25
The system should allow maintenance staff to view equipment associated with maintenance records.

FR-26
The system should associate a maintenance record with the relevant Equipment ID.

FR-27
The system should allow maintenance staff to update repair status.

FR-28
The system should support marking equipment as available after repair and inspection.

3.2.6 Administrator Functions

The following requirements describe planned functionality.

FR-29
The system should provide an administrator with functionality for managing user records.

FR-30
The system should support management of user roles and permissions.

FR-31
The system should provide appropriate access restrictions for administrative operations.

3.2.7 Feedback and Notifications

These requirements describe planned functionality.

FR-32
The system should allow users to submit feedback, suggestions, ratings, or complaints.

FR-33
The system should store feedback records for later review by authorized personnel.

FR-34
The system should support notifications for relevant events such as booking updates and equipment-related status changes.

FR-35
The system should maintain notification read status where notification functionality is implemented.

3.3 Equipment Request Workflow

The equipment request process begins when a student or coach views the equipment catalogue and selects an available item. The application checks whether sufficient equipment is available. If availability is confirmed, the user submits the request directly without waiting for Sports Officer approval. The request is recorded in the system, and the equipment can be issued according to the implemented resource management process.

If the requested equipment is unavailable, the application should inform the user that the item cannot currently be requested.

3.4 Facility Booking Workflow

The facility booking process begins when a student or coach selects a facility and views its available time slots. The user selects a date and time and submits the booking request. The system checks the existing booking records for conflicts.

If the selected slot is available, the system records and confirms the booking automatically. No separate Sports Officer approval is required for a normal booking. If the slot conflicts with an existing booking, the system informs the user that the selected slot is unavailable and allows the user to choose another slot.

3.5 Maintenance Workflow

The intended maintenance process begins when an equipment item is reported as requiring attention. Maintenance staff identify the item using its Equipment ID and inspect the reported problem. If repair is needed, the maintenance status is updated while the item is being repaired. After repair, the equipment is inspected again and its status is updated. Equipment should be marked available only when it is considered suitable for use.

The maintenance interface and complete tracking workflow remain under development.

3.6 Activity Diagram

The Activity Diagram will represent a major system workflow, such as facility booking or equipment requests. It will include user actions, availability checking, decision points, and the outcomes of successful and unsuccessful operations.

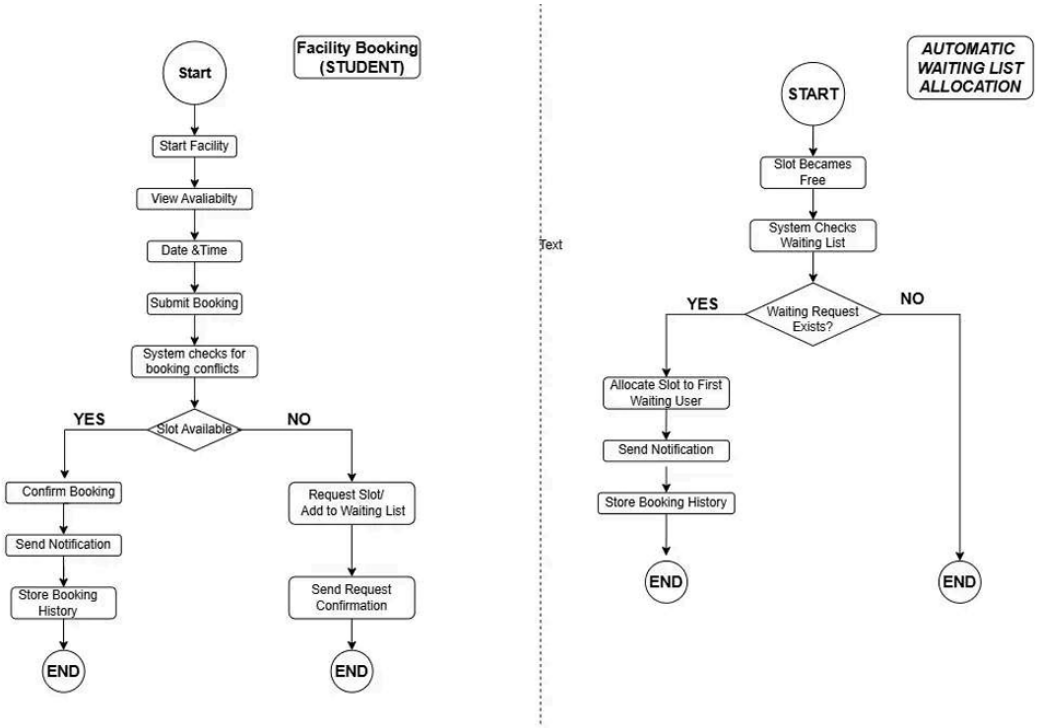


Figure 3.1: Activity Diagram

3.7 Class Diagram

The Class Diagram will represent the main entities and their relationships. The principal classes include User, Equipment, Facility, Equipment Request, Booking, Maintenance, Feedback, and Notification.

The diagram will show relevant attributes and associations, including the relationship between users and bookings, facilities and bookings, equipment and equipment requests, and equipment and maintenance records.

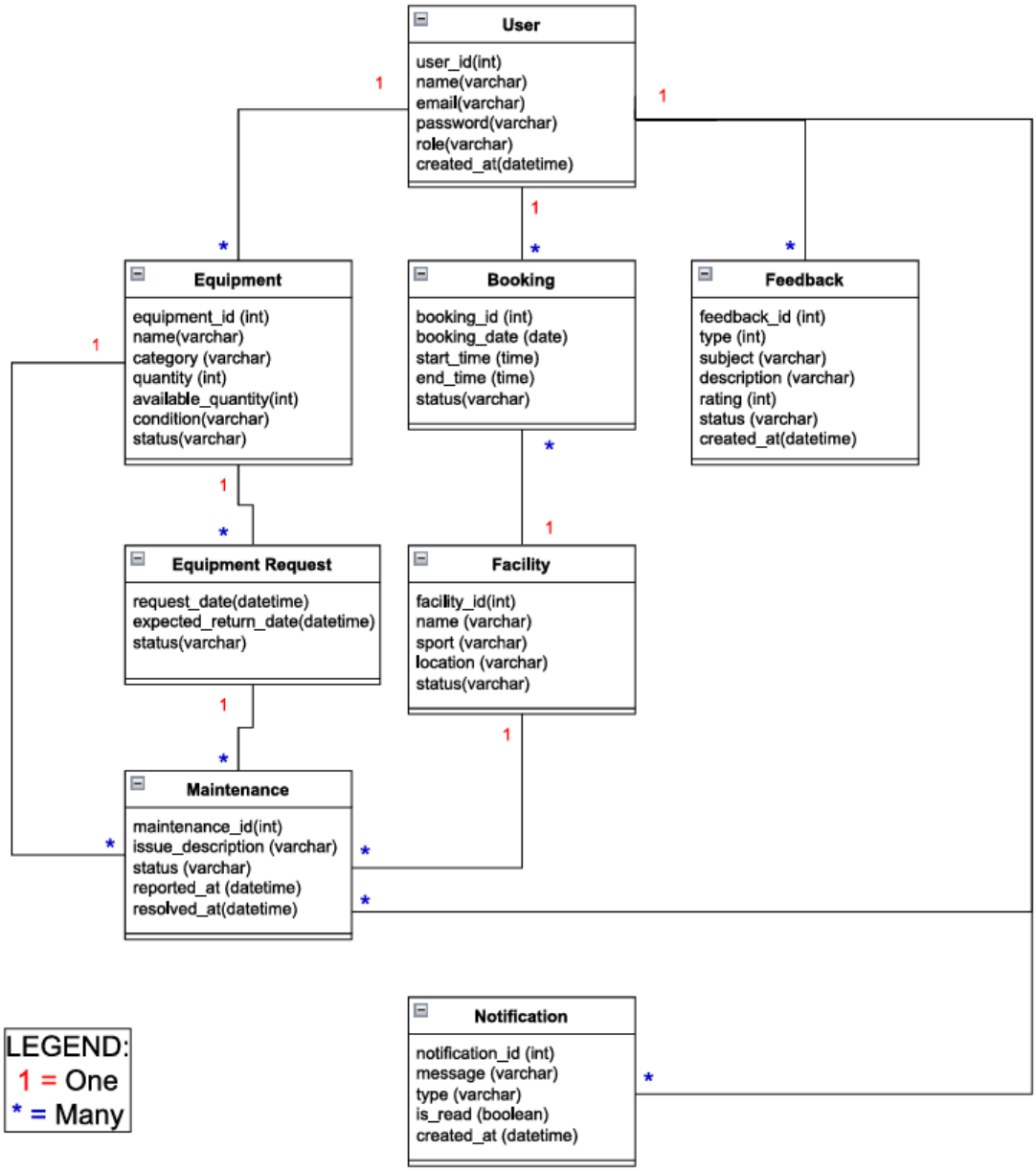


Figure 3.2: Class Diagram

4 Other Nonfunctional Requirements

4.1 Performance Requirements

NFR-01
The application should display normal resource information within a reasonable response time under ordinary prototype conditions.

NFR-02
The backend should process equipment requests and booking operations without unnecessary delays.

NFR-03
The system should check booking availability before confirming a booking.

NFR-04
The application should provide visible feedback when an operation is loading, completed, or unsuccessful.

No numerical performance targets are claimed for the current prototype. Such targets can be established after testing on representative devices and under defined operating conditions.

4.2 Security Requirements

NFR-05
The system shall require users to authenticate before accessing protected functionality.

NFR-06
The system shall restrict operations according to the user's role.

NFR-07
Passwords shall not be stored as plain text in a production-ready implementation.

NFR-08
The backend shall validate incoming data before performing database operations.

NFR-09
Database credentials and other sensitive configuration values should not be embedded in publicly shared source code.

NFR-10
The system should prevent unauthorized users from modifying equipment, facilities, or booking records.

These requirements define the intended security properties. Their implementation and verification must be assessed separately during testing.

4.3 Reliability and Data Integrity

NFR-11
The system shall store core records in PostgreSQL so that data can be retrieved after the application is restarted.

NFR-12
The system shall validate booking availability before confirming a booking.

NFR-13
The system should maintain valid relationships between users, equipment, facilities, bookings, and requests.

NFR-14
Failed operations should provide an appropriate error message rather than silently reporting success.

NFR-15
Equipment availability information should be updated consistently when equipment-related operations are completed.

4.4 Usability Requirements

NFR-16
The mobile interface should use clear labels and understandable navigation.

NFR-17
Common operations such as viewing facilities, booking a slot, and requesting equipment should require a reasonable number of steps.

NFR-18
Error messages should explain the problem in language understandable to ordinary users.

NFR-19
The interface should distinguish available, unavailable, and under-maintenance resources where the corresponding information is implemented.

4.5 Maintainability

NFR-20
The application should keep mobile interface code, backend logic, and database operations organized into understandable components.

NFR-21
The system should allow additional modules to be introduced without unnecessarily rewriting existing core features.

NFR-22
Source code should be maintained through Git and GitHub to support version tracking and team collaboration.

4.6 Portability and Compatibility

NFR-23
The Flutter application should be developed for supported target devices and platforms selected by the project team.

NFR-24
The mobile interface should remain usable on supported screen sizes.

NFR-25
The backend should communicate with the database using the configured PostgreSQL connection.

4.7 Software Quality Attributes

The principal quality attributes of the system are:

- **Correctness:** The application should perform the specified equipment and booking operations correctly.
- **Usability:** Users should be able to understand the mobile interface and complete common tasks.
- **Reliability:** The system should handle normal operations and errors without incorrectly reporting success.
- **Security:** Protected functionality should be available only to authorized users.
- **Maintainability:** The codebase should support debugging, corrections, and further development.
- **Consistency:** Equipment availability and booking information should agree with the stored records.

4.8 Business Rules

The system follows these business rules:

1. Users must access protected functionality through the login process.
2. Students and coaches can request available equipment directly without a separate Sports Officer approval step.
3. Facility bookings are automatically confirmed only when the requested slot is available.
4. A facility should not have two confirmed bookings for overlapping time periods.
5. Equipment marked as unavailable or under maintenance should not be presented as normally available.
6. Each equipment record should have an identifier that allows it to be distinguished from other equipment records.
7. Role-specific operations should be restricted to authorized users.

5 Other Requirements

5.1 Database Requirements

The PostgreSQL database is intended to store the principal entities required by the system. These include:

- Users
- Equipment
- Facilities
- Equipment Requests
- Bookings
- Maintenance Records
- Feedback
- Notifications

The database schema should maintain appropriate identifiers and relationships between entities. Equipment maintenance records should reference the relevant Equipment ID. Booking records should identify the user and facility involved, along with the requested date, time, and status.

5.2 External Interface Requirements

5.2.1 Mobile User Interface

The mobile application shall provide interfaces for role selection, login, equipment browsing, equipment requests, facility browsing, and facility booking, according to the functionality implemented for each role.

5.2.2 Backend Interface

The mobile application communicates with the Node.js backend through application programming interfaces. The backend processes requests and interacts with PostgreSQL.

5.2.3 Database Interface

The backend uses a PostgreSQL connection to store and retrieve system records. Database credentials should be managed through configuration rather than committed as plain text in a public repository.

5.3 Project Constraints

The project is being developed as a university software engineering prototype. The available development time and team resources limit the number of modules that can be completed during the initial stage.

The current scope prioritizes the working mobile application, core equipment and facility operations, booking, equipment requests, and backend/database integration.

Administrator functionality, maintenance tracking, feedback, and notifications require further implementation. The project does not claim production deployment, complete security certification, or formal performance results at this stage.

5.4 Future Enhancements

Potential future enhancements include:

- Complete administrator functionality.
- Complete maintenance staff interface and repair tracking.
- Equipment damage reporting.
- Feedback and complaint management.
- Notifications for booking and equipment updates.
- Improved booking and equipment history.
- More detailed usage and resource reports.
- Expanded testing and usability evaluation.

5.5 Conclusion

The Sports Resource Management System is a mobile application designed to organize university sports equipment and facility management.

This SRS defines the intended users, system scope, objectives, major features, functional requirements, non-functional requirements, constraints, and future extensions. It establishes a consistent reference for implementing and evaluating the project.

The current prototype prioritizes the core equipment and facility workflows. Facility bookings are automatically confirmed when the selected slot is available, and students or coaches can directly request available equipment without a separate approval step. Administrator, maintenance, feedback, and notification modules remain planned extensions until their functionality is implemented and tested.

The document may be updated as the prototype evolves and additional requirements are clarified.